# Mileage forecast + erroneous-report flags

One prediction table for all 656,601 test rows: forecast, cold start and error flags

**1,406 km**

**Forecast MAE, 133 days**

mean over 3 cut-off dates,
held-out vehicles

**590-666 km**

**Nowcast MAE, up to 28 days**

10-13% better than the best rule

**99.3%**

**Recall on injected errors**

5,338 of 493,554 reports flagged (1.1%)

**163,047**

**forecast rows filled**

159,302 model + 3,745 cold start
(149 new cars without history)

**0 / 1**

**error flag per reported mileage**

rule violations + deviation from
expected usage, with score and reason

**1 file**

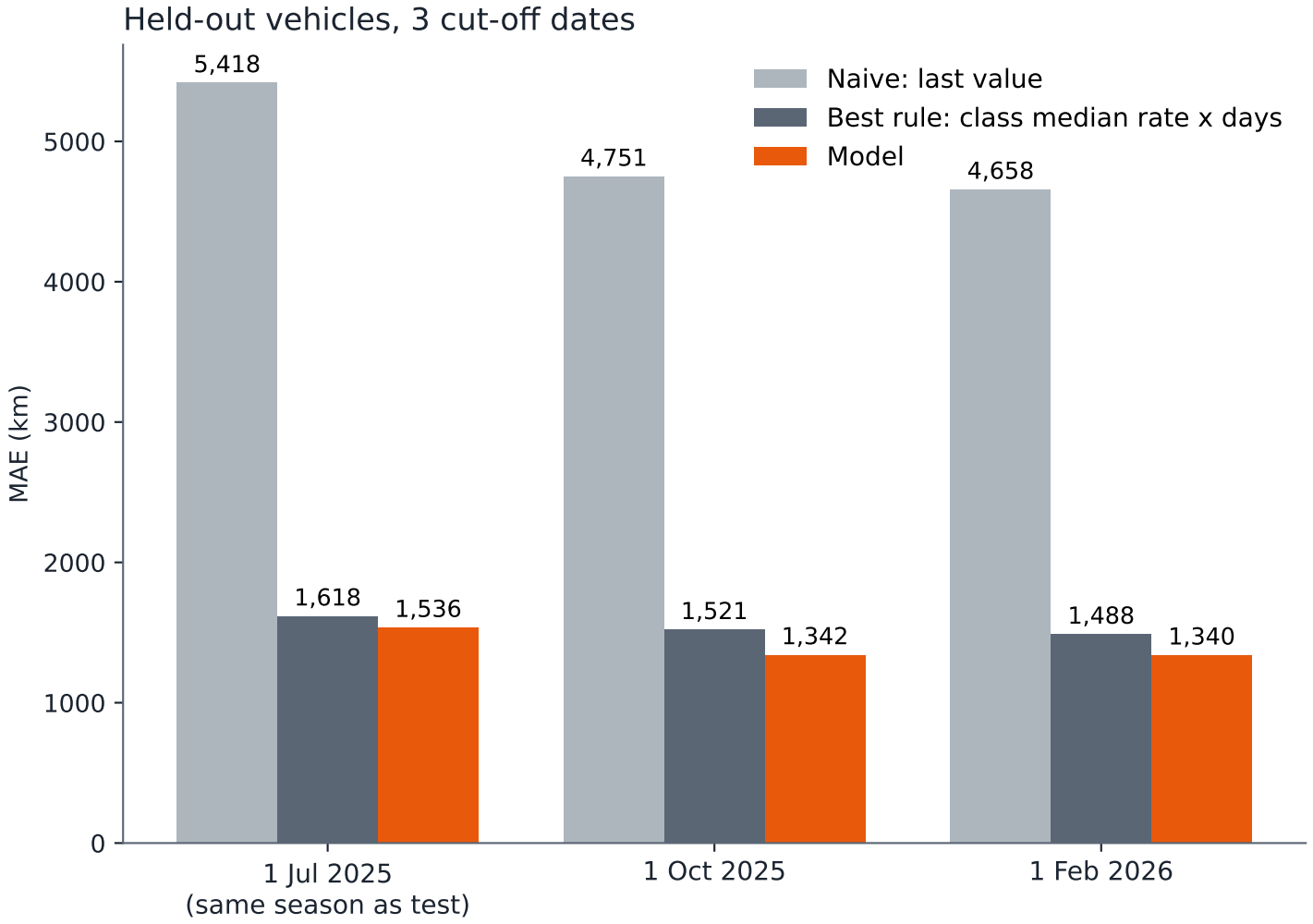
**submission/predictions.parquet**

same row order as test_data.parquet,
key: row_id

All accuracy numbers are measured on vehicles held out from training and tuning (no labels exist for the hidden test period).

# Forecast: 133 days ahead

Gradient boosting on the mileage increment since the last known reading; error in km (MAE), lower is better

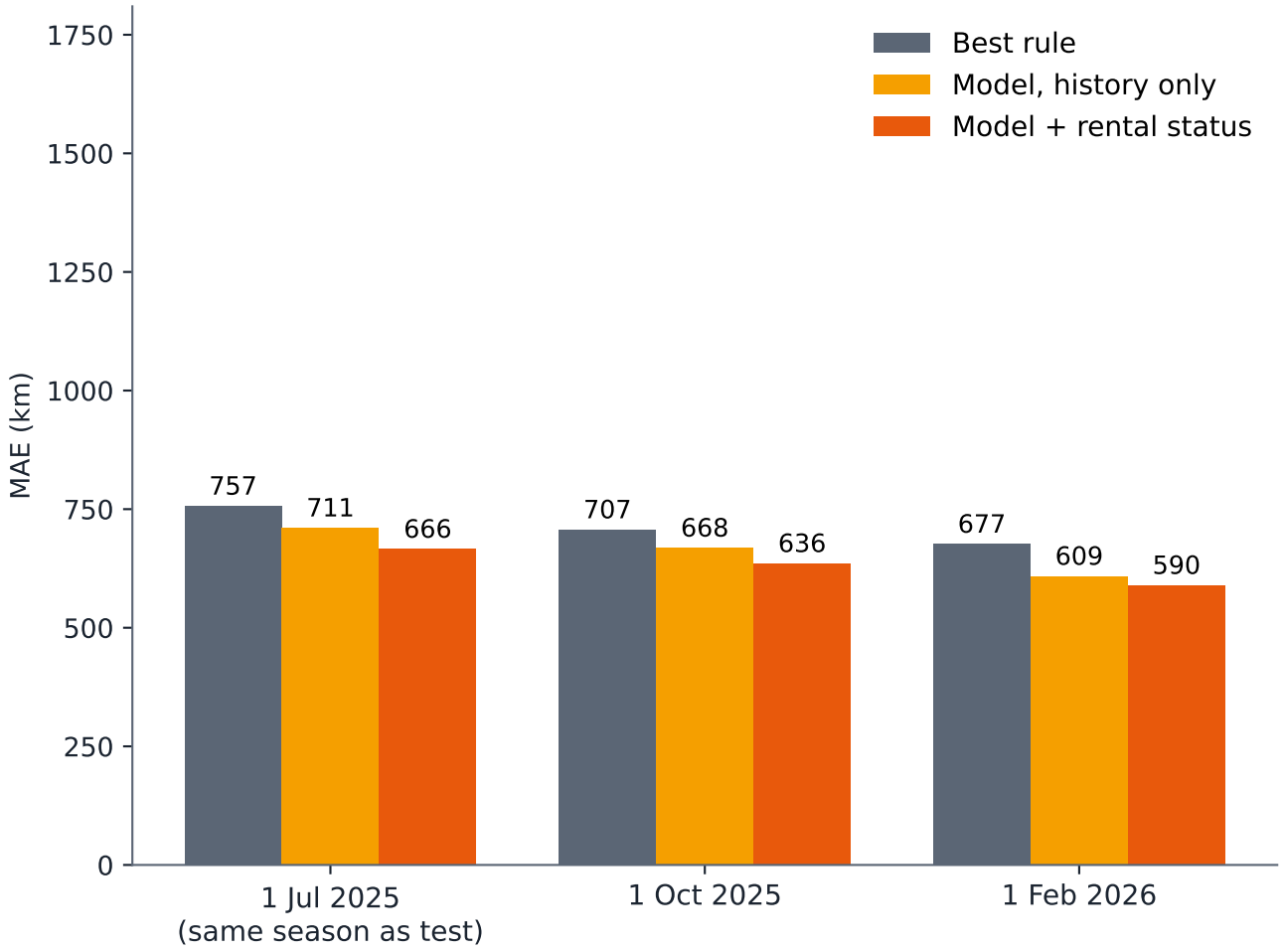



Model beats the best rule by

| | |
|---|---|
| **5.0%** | 2025-07-01 |
| **11.8%** | 2025-10-01 |
| **10.0%** | 2026-02-01 |

# Nowcast and new vehicles

Nowcast: mileage on a day without report (up to 28 days), using the rental status of that day




## Cold start: 149 new cars with no mileage history

start at a typical new-car reading (about 12 km) and follow the median growth curve of new vehicles by age in days

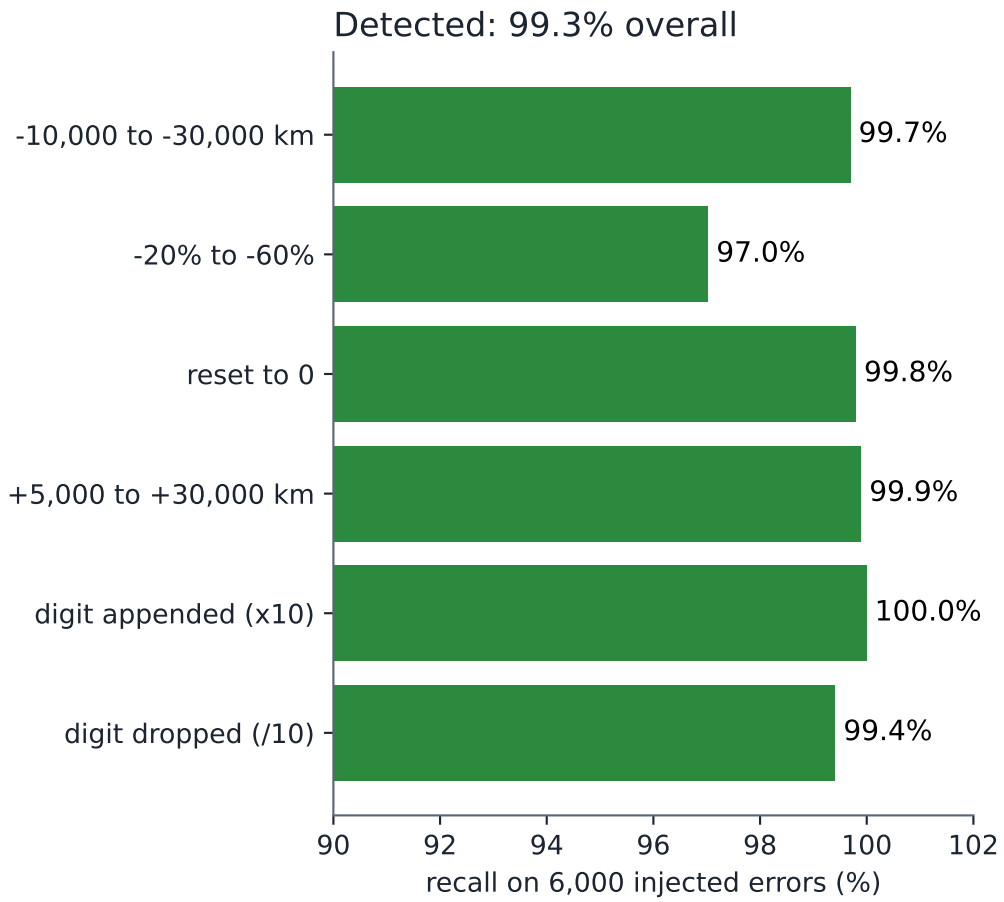
**MAE on simulated new cars (km)**

| cut-off | fleet rate | age curve |
|---|---|---|
| 2025-07-01 | 1,312 | **1,255** |
| 2025-10-01 | 1,091 | **1,091** |
| 2026-02-01 | 1,137 | **961** |

The age curve is never worse than the fleet-rate rule and cuts the error by 15% at one cut-off.

# Erroneous-report flags

A report is flagged when it breaks physics or deviates strongly from expected usage




## Five flags, combined with OR

rollback: reading below the previous one (longest non-decreasing subsequence marks the outlier itself)

too high: above all later readings

speed: more than 1,500 km/day between readings

high / low: increase far above or below the expected usage for that time gap (quantile bands per gap)

## On the test file

5,338 of 493,554 reports flagged (1.1%)

2,781 by rules only, 2,257 by deviation only, 300 by both

every flag has a score and a plain-text reason

Injected errors are large; subtle manipulation is not covered by this test. No ground-truth labels exist.

# Deliverables, limits, next steps

### What is in the repo

submission/predictions.parquet: forecast + error flag for every test row

models/: final forecast model and nowcast model (joblib)

results/: all evaluation tables; reports/figures/: feature importance

src/ + scripts/: full pipeline, run with python -m scripts.run_<step>

SUBMISSION.md: approach, results, how to run

### Next steps

Seasonal index and fleet trend as forecast features

Out-of-fold nowcast as expected value for the flags,
with quantile uncertainty bands

Cross-validation over vehicles

Checks with subtle manipulation (a few percent)

### Known limits

Model underestimates in the summer cut-off (bias -516 km): no seasonal feature yet

Gain over the best rule varies by holdout and date: 5% to 12%

Error flags use a simple expected rate, not the trained model; no labels to verify precision

Nowcast is not filled per test row (empty test rows have no rental status)